

BounceCoach: Real-Time Basketball Dribble Training with On-Site Large Language Model Guidance

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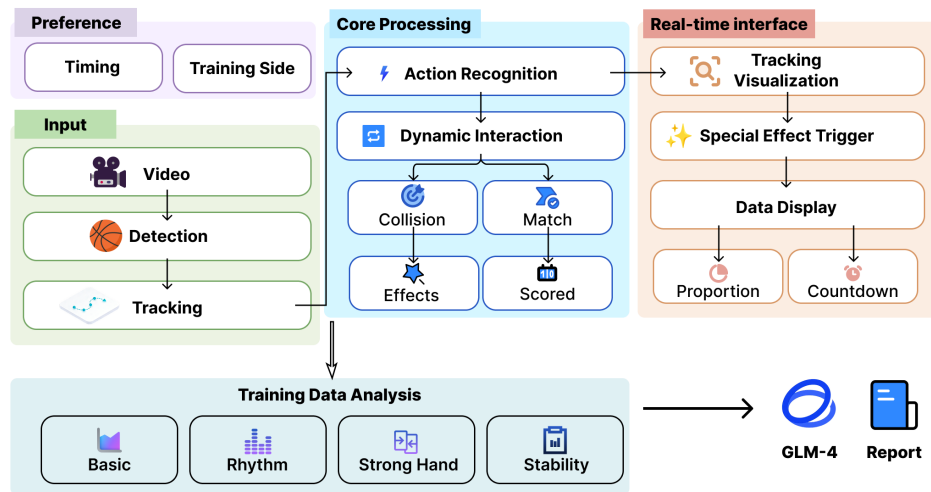


Figure 1: Overview of our BounceCoach application.

ABSTRACT

We present BounceCoach, an edge-based Augmented Reality (AR) training application that turns casual basketball dribble practice into an engaging, AI-guided game. By fusing low-latency on-device pose tracking with an on-site distilled large language model, the framework delivers instant verbal form cues, rhythm tips and mini-challenges while the user bounces to music. Edge offloading keeps motion-to-feedback below 40 ms, maintaining immersion and safety. A 24-subject pilot shows 28% gain in dribble speed consistency and $1.6\times$ higher self-reported fun versus non-AR training, validating the approach for lightweight, location-free sports entertainment.

Index Terms: AI coach, Fun drills, AR basketball.

1 INTRODUCTION

Dribbling is a cornerstone of basketball training, yet traditional instruction still hinges on coach experience and athletes' subjective self-perception [1, 4]. This paradigm suffers from three inherent shortcomings: (1) athletes practicing alone struggle to detect and correct subtle kinematic deviations; (2) performance evaluation remains largely subjective, lacking objective and quantifiable metrics;

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and (3) high-quality feedback is tightly coupled to costly coaching resources, impeding large-scale deployment.

Although high-precision motion capture systems can reconstruct movements at millimeter-level fidelity, their prohibitive cost places them beyond the reach of everyday athletes. Conversely, consumer-grade devices such as smart basketballs readily output multi-dimensional mechanical parameters [3]. However, they fail to capture spatial trajectories, offering only scant support for fine-grained ball-control training.

The essence of dribble training is to improve athletes' ball-control stability under varying conditions and to enable them to immediate, on-the-fly adjustments. Therefore, an ideal digital system must track the ball's position within a defined space in real time and reconfigure its parameters according to the athlete's performance, accommodating individual skill levels and training objectives.

We present BounceCoach, an edge-based AR dribble-training framework that integrates real-time ball tracking, adaptive target-point generation within the training space, and post-session intelligent feedback (as shown in Fig. 1). The system defines a training zone around the user and continuously records the ball's position and state. From the tracked spatio-temporal features, BounceCoach computes on-the-fly statistics, visualizes them, and guides the user to adjust dribble rhythm and improve spatial control. After training, a lightweight Large Language Model (LLM) analyzes the collected data and pinpoints weaknesses. Compared with laboratory-grade motion-capture setups, BounceCoach utilizes regular cameras and edge devices, balancing cost and portability for sustainable use. The main contributions are as follows:

- An edge-based AR dribbling training system that promotes

spatial control through interactive virtual targets.

- A low-cost and lightweight alternative to professional motion capture for everyday basketball practice.
- An AI-based coaching approach that converts training data into interpretable and personalized feedback.

2 DESIGN AND IMPLEMENTATION

2.1 System Implementation

Real-Time Ball Tracking and Spatial Feedback. BounceCoach employs a vision-based ball tracking pipeline to support real-time spatial feedback. A lightweight object detection model continuously identifies the basketball and estimates its screen-space position, which is temporally buffered to form a short motion history. This trajectory is rendered as a dynamic visual trace, allowing users to intuitively perceive ball motion continuity and spatial deviations during dribbling. To maintain stable feedback under noisy detection conditions, a smoothing strategy is applied to suppress sudden positional jumps, ensuring that visual feedback remains coherent and supportive of integrative motor learning [2].

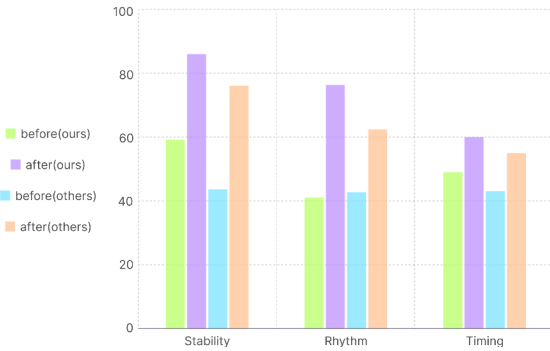


Figure 2: Comparison of dribbling performance with and without BounceCoach. Users trained with BounceCoach show larger improvements in stability, rhythm, and timing after training, compared to baseline performance and to users practicing without the system.

Interactive Targets and Visual Reinforcement. Virtual training targets are rendered as circular regions within the user's interaction space. Ball-target interactions are determined through real-time spatial proximity checks, triggering both score updates and visual effects upon successful hits. Particle-based visual feedback is used to reinforce successful actions, enhancing engagement and action-perception coupling. Training statistics, including left and right hand dribbling frequency, are visualized using dynamically scaled charts. These visualizations update in real time, providing immediate and interpretable feedback on hand dominance and rhythm balance.

LLM-Driven Training Feedback. To address the lack of objective evaluation in traditional dribbling practice, BounceCoach integrates a large language model for post-session performance analysis. After each session, the model processes key metrics (dribble count, score, scoring rate) and converts them into feedback parameters, enabling a virtual coach to deliver personalized guidance in natural language.

As shown in Fig. 2, users who train with BounceCoach consistently demonstrate greater improvements in stability, rhythm, and timing compared to those practicing without the system. These quantitative gains indicate that LLM-driven reflection effectively reinforces real-time embodied feedback by providing higher-level interpretation and motivation, bridging low-level performance metrics with human-understandable coaching insights.

2.2 Training Interaction Design

BounceCoach is designed as an embodied, real-time dribbling training experience that supports lightweight and self-guided practice. Users can select the training hand and session duration, with different durations corresponding to distinct rhythmic patterns and difficulty levels.

As shown in Fig. 3, the system continuously generates virtual target regions within the user's reachable space. Users score points by guiding the basketball into the highlighted region within a limited time window, encouraging precise spatial control and rhythm regulation. This interaction loop, which includes perceiving targets, adjusting movements, and receiving feedback, forms a complete closed-loop training system. The objective is to maximize the cumulative score within the designated training duration.

After each session, BounceCoach forwards the statistics to the LLM (GLM-4), which autonomously analyzes the results and produces a concise evaluation report, allowing users to review their training outcomes both visually and quantitatively.

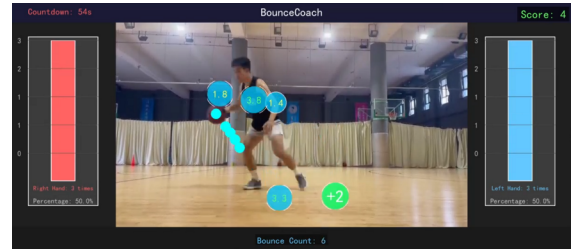


Figure 3: System interfere of BounceCoach.

3 CONCLUSION

This work presents BounceCoach, a lightweight and accessible AR-inspired basketball dribbling training system that transforms casual practice into an embodied, data-driven learning experience. By combining real-time vision-based ball tracking, interactive spatial targets, and AI-driven post-session feedback, BounceCoach supports both immediate motor correction and reflective skill improvement. A pilot study indicates the system improves dribbling consistency and engagement, showing potential for scalable, location-free sports training and entertainment.

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